

Data from: **The functional diversity of retinal ganglion cells in the mouse**

This dataset contains raw and processed data acquired using 2-photon imaging with OGB-1 of mouse retinal ganglion cells using a standardized set of stimuli as described in Baden et al. (2016). Visualization functions for use with MATLAB are available from www.retinal-functomics.org.

Variable name	Dim 1	Dim 2	Dim 3	Dim 4	Remarks
cluster_idx	#cells				Cluster numbers as in Suppl. Data S3; RGCs: 1-39, uncertain: 40-49, ACs: 50-75
group_idx	#cells				Group numbers as in Suppl. Data S4; RGCs: 1-28, uncertain: 29-32, ACs: 33-46
sel_idx	#cells				Indicates whether cells meet the selection criterion for use in clustering
cell_id	#cells	7			year / month / day / experiment number / global scan number / scan number / cell number
cell_area	#cells				ROI area
cell_volume	#cells				Volume of cell body
cell_dsi	#cells				Direction selectivity index
cell_osi	#cells				Orientation selectivity index
cell_dp	#cells				Direction tuning p-value
cell_op	#cells				Orientation tuning p-value
cell_ooi	#cells				ON-OFF-index
cell_ffi	#cells				Full-Field-index
chirp_avg	#samples	#cells			Median of all repeats, scaled to $\max(\text{abs}(x))=1$
chirp_byrepeat	#samples	#repeats	#cells		Individual repeats, scaled with same scaling factor as chirp_avg
chirp_time	#samples				Time base for responses
chirp_qi					Quality index as defined in paper
chirp_stim	#samples				Stimulus waveform of chirp
chirp_stim_time	#samples				Time base for stimulus waveform
bar_tc	#samples	#cells			Extracted time course from bar stimulus (SVD on average across directions)
bar_byrepeat	#samples	#dirs	#repeats	#cells	Individual repeats
bar_time	#samples				Time base of bar stimulus
bar_qi	#cells				Quality index as defined in paper
color_avg	#samples	#cells			Median of all repeats, scaled to $\max(\text{abs}(x))=1$
color_byrepeat	#samples	#repeats	#cells		Individual repeats, scaled with same scaling factor as chirp_avg
color_time	#samples				Time base for responses

color_qi	#cells			Quality index as defined in paper
noise_raw	#samples	#cells		Response to checkerboard noise stimulus
noise_time	#samples			Time base for response to checkerboard noise stimulus
noise_stim	#samples			Checkerboard noise stimulus
noise_stim_time	#samples			Time base for checkerboard noise stimulus
rf_tc	#samples	#cells		Time kernel of RF (at 10 times the stimulus frequency)
rf_map		20	15 #cells	Spatial map of RF
rf_gauss_mean	#cells		2	2D mean of gauss fit to the spatial map
rf_gauss_std	#cells		2	2D std of gauss fit to the spatial map
rf_size	#cells			2-std-ellipse area as receptive field size
rf_time	#samples			Time of RF time kernel
rf_qi	#cells			Quality index as defined in paper
immuno_gad	#cells			1: staining present; 0.75: staining likely present; 0.5: staining undecided; 0.25: staining likely not present; 0: staining not present; NaN: not stained
immuno_chat	#cells			
immuno_smi32	#cells			
immuno_melanopsin	#cells			
genetics_pv	#cells			1: genetic marker present; 0: genetic marker not present
genetics_pcp	#cells			
structural		64	64 #scans	structural scan of each recorded field (indexed by global scan number)
rois		64	64 #scans	rois used for each recorded field (indexed by global scan number)

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